Practical QML in the NISQ Era: From Static Encoding to Automated Hybrid Pipelines

Sarthak Shaurya (22116082)
Aryan Chaudhary (22115031)
Nitesh Thalor (22116062)

Department of Electronics & Communication Engineering
Indian Institute of Technology Roorkee

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Problem Formulation: The NISQ Bottleneck

The Hardware Challenge

Current Noisy Intermediate-Scale Quantum (NISQ) devices lack Quantum RAM. Processing datasets natively is heavily constrained by limited qubit budgets and high gate error rates.

Baseline Proof-of-Concept (Step 1)

- **Approach:** Implemented a linear-time $\log_2 N$ encoding technique for classical-to-quantum embedding.
- **Execution:** Successfully ran a simplified Quantum-Kernel Support Vector Machine (sqKSVM).
- **Limitation:** While feasible, global encoding restricts this baseline to small, static, tabular clinical datasets.

Problem Formulation: The High-Dimensional Data Problem

The Scaling Limitation

Real-world signals, such as continuous audio and speech, are massive and temporally complex. Attempting global quantum encoding of raw audio waveforms instantly exceeds current NISQ qubit limits.

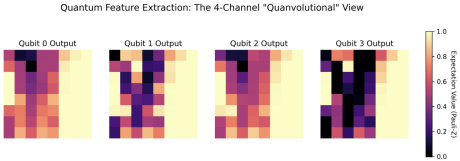
Our Engineering Objective

- **The Question:** How do we scale quantum feature extraction to high-dimensional, temporal data without requiring deeper, error-prone quantum circuits?
- **The Pivot:** Moving from static clinical data to dynamic Automatic Speech Recognition (ASR).

Technique Explanation: Decentralizing Extraction (Step 2)

Quantum Convolutional Neural Networks (QCNN)

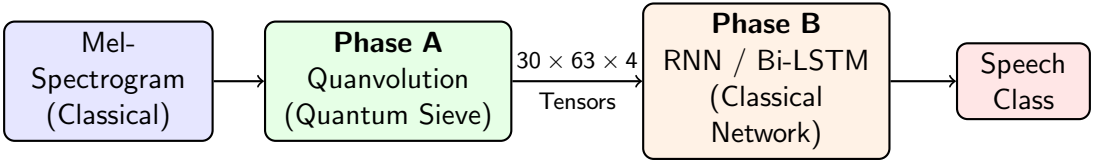
- **Shift in Architecture:** Replaced global data encoding with localized "Quanvolutional" filters.
- **Data Transformation:** Converted raw audio signals into 2D Mel-spectrograms.
- **Execution:** A parameterized quantum circuit acts as a sliding window, sweeping across the acoustic spectrogram to extract entangled features.



(Fig 1: Quanvolution Sliding Window)

Visualizes the 4 distinct 8x8 spatial feature maps (Pauli-Z expectation values) extracted by the 4 qubits.

Technique Explanation: The Hybrid CQ Architecture



Architectural Advantage

The QPU acts purely as a non-linear feature extractor. The extracted tensors are collapsed classically and fed into an RNN. This perfectly respects NISQ hardware constraints by maintaining shallow circuit depth.

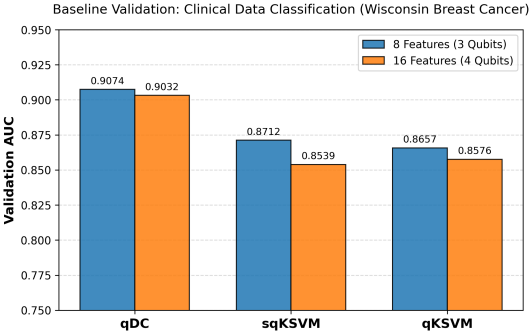
Results: Step 1 Baseline Classification

Clinical sqKSVM Metrics

- **Dataset:** Tabular clinical records.
- **Encoding:** Linear-time $\log_2 N$ state preparation.
- **Outcome:** Successfully classified data points utilizing a strict qubit budget.

Validation

Proves the feasibility of mathematical data encoding on constrained hardware.



(Fig 2: sqKSVM Classification Results)

Demonstrates successful binary classification of clinical data boundaries using a strict quantum kernel.

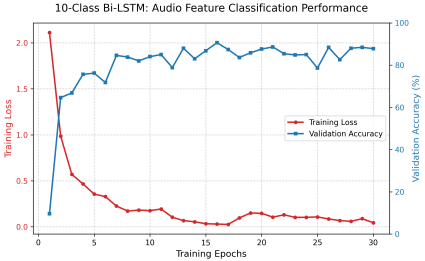
Results: Step 2 Benchmark Replication (Cloud)

10-Class Sequence Training (Keras Bi-LSTM)

Processed the full 3,000-file dataset through the Bi-LSTM sequence architecture, relying entirely on the extracted quantum spatial tensors to successfully replicate the published baseline.

Training Metric	Achieved Value
Total Parameters	180,437
Dataset Classes	10 (Full Audio)
Peak Validation Epoch	Epoch 16
Training Accuracy	99.07%
Validation Accuracy	**90.60%**

Table 1: Final empirical 10-class results.



(Fig 3: Bi-LSTM Epoch Metrics)

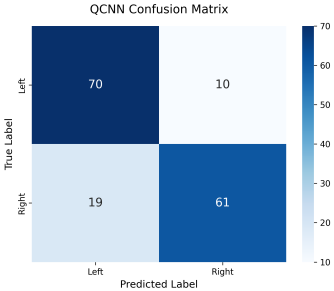
Model convergence hitting peak val. acc. at Epoch 16.

Results: Step 2 Local Hardware Validation (PyTorch)

Engineering Prototype: Local Hardware Validation

Prior to executing the computationally expensive 10-class extraction on the Linux server, the quantum spatial features were mathematically validated locally.

- **Subset Testing:** Processed a binary audio subset (Left vs. Right).
- **Architecture:** Built a custom, lightweight PyTorch CNN to ingest the 4-channel quantum tensors.
- **Outcome:** Confirmed strong feature separation, proving the quantum circuit's validity before cloud scaling.



(Fig 4: Binary Confusion Matrix)

Proves strong subset separation utilizing the custom

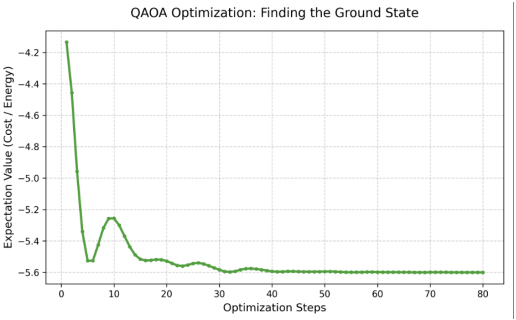
Novelty: Automating the Pipeline with QAOA (Step 3)

The Operational Friction

Manually tuning hyperparameters across a delegated CPU/QPU architecture is unscalable for production.

The QAOA Solution

- Formulated Hyperparameter Optimization as a QUBO problem.
- Deployed the **Quantum Approximate Optimization Algorithm (QAOA)** to navigate the parameter space autonomously.



(Fig 5: QAOA Optimizer Convergence)

Illustrates the QAOA solver dynamically minimizing the cost function.

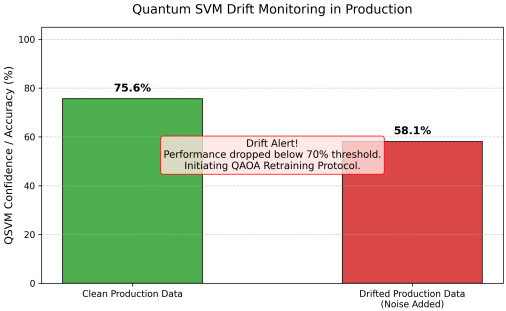
Novelty: Closing the Loop with Production Monitoring

Repurposing the Baseline

We brought back the **Quantum SVM** from Step 1, completely shifting its role. Instead of clinical classification, it now acts as a real-time boundary monitor to detect data drift in live audio streams.

Final Automated Architecture

1. **QCNN + RNN:** Acoustic transcription.
2. **QAOA:** Continuous optimization.
3. **QSVM:** Instant drift flagging.



(Fig 6: QSVM Production Drift Detection)

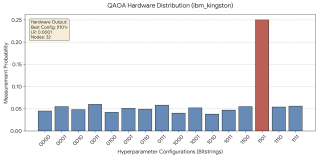
Displays the QSVM boundary successfully flagging out-of-distribution audio samples.

Hardware Validation: IBM QPU Execution

Executing on Real Quantum Hardware

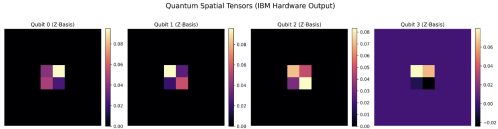
To validate the hybrid pipeline on actual NISQ devices, aggressive optimizations were deployed to survive queue times and hardware coherence limits.

QAOA Optimization (ibm_kingston)



- **Hardware Adaptation:** Reduced optimization steps to 2 and circuit depth to 1 to mitigate high decoherence rates.
- **Execution:** Successfully executed the classical-quantum feedback loop on the `ibm_kingston` device.

QCNN Spatial Tensors



- **Center-Crop Strategy:** Isolated the center 4×4 frequency patch to drop the workload from 64 to exactly 4 quantum circuits.
- **Safety Padding:** Embedded the results into an 8×8 zero matrix to maintain downstream shape compatibility.

Summary & References

Project Summary

1. **Feasibility:** Validated NISQ constraints using linear-time encoding (sqKSVM).
2. **Scalability:** Scaled to ASR using local QCNN feature extraction combined with classical RNNs.
3. **Deployability (Novelty):** Automated the entire hybrid CQ pipeline using QAOA for HPO and QSVM for drift monitoring.

Key References

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3. E. Farhi, J. Goldstone, and S. Gutmann, "A Quantum Approximate Optimization Algorithm," *arXiv preprint arXiv:1411.4028*, 2014.
4. C.-H. H. Yang, J. Qi, S. M. Siniscalchi, et al., "Decentralizing Feature Extraction with Quantum Convolutional Neural Network for Automatic Speech Recognition," *arXiv:2010.13309*, 2021.